

QVerse — Week 3: One Qubit — States, X, H, Z, and Measurement

Theory Booklet | Introduction to Quantum Computing & Programming

QVerse

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How this booklet fits the course

This theory booklet is designed to be read alongside `03_one_qubit_states_and_gates.ipynb`. It contains the conceptual and mathematical material that should be explained during Week 3; the notebook then turns those ideas into experiments, circuit construction, and exercises.

Learning objectives

- Prepare $|0\rangle$, $|1\rangle$, $|+\rangle$, and $|-\rangle$.
- Use X, H, and Z gates.
- Inspect statevectors and sampled measurements.
- Distinguish state information from measurement outcomes.

Recommended workflow

1. Read the relevant theory section.
2. Predict the result of the corresponding notebook experiment.
3. Run the notebook.
4. Explain any difference between the prediction and the output.
5. Complete the core exercises without treating the solution notebook as a substitute for reasoning.

1. Four one-qubit states to know

The computational-basis states are

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Two equally important superposition states are

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

They satisfy

$$H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle.$$

The pair $\{|+\rangle, |-\rangle\}$ forms the **X basis**, while $\{|0\rangle, |1\rangle\}$ forms the **Z basis**.

2. Three essential gates: X, Z, and H

The Pauli-X gate is

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

and flips computational-basis states.

The Pauli-Z gate is

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

It leaves $|0\rangle$ unchanged and adds a minus sign to $|1\rangle$:

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

The Hadamard gate is

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

It converts between the Z and X bases:

$$H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle,$$

$$H|+\rangle = |0\rangle, \quad H|-\rangle = |1\rangle.$$

This basis-change interpretation is more useful than memorizing the matrix.

3. Superposition does not mean uncertainty in every basis

Consider

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

If measured in the Z basis,

$$P(0) = P(1) = \frac{1}{2}.$$

But $|+\rangle$ is an eigenstate of X , so a measurement in the X basis is deterministic.

This illustrates a fundamental point: a state is not “random” in an absolute sense. Measurement statistics depend on the chosen basis.

4. How to measure in another basis

Hardware measurements are commonly expressed in the computational basis. To perform an X-basis measurement, rotate the X basis into the Z basis before measurement.

Because

$$H|+\rangle = |0\rangle, \quad H|-\rangle = |1\rangle,$$

an X-basis measurement can be implemented as:

1. apply H ;
2. measure in the Z basis.

This pattern generalizes: **basis change + computational measurement** is a common way to realize other measurements.

5. Statevector versus sampled counts

A pure one-qubit statevector contains complex amplitudes:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

It tells us more than one set of sampled counts. For example, the states

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}},$$

and

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

produce identical Z-basis probabilities:

$$P(0) = P(1) = \frac{1}{2}.$$

However, they behave differently when another gate is applied before measurement. The relative sign is physically meaningful because it changes interference.

Therefore, matching one histogram does not prove that two quantum states are identical.

6. The Z gate is a phase flip

Applying Z to $|+\rangle$ gives

$$Z|+\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle.$$

The Z-basis probabilities remain unchanged:

$$\left| \frac{1}{\sqrt{2}} \right|^2 = \left| -\frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

So the effect of Z is invisible if we immediately measure in the Z basis. It becomes visible after a suitable basis change.

This is the first clear example of hidden phase information becoming observable through interference.

7. H is its own inverse

Because

$$H^2 = I,$$

two consecutive Hadamard gates cancel:

$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{H} |0\rangle.$$

Similarly,

$$|1\rangle \xrightarrow{H} |-\rangle \xrightarrow{H} |1\rangle.$$

This is useful both conceptually and computationally: a circuit can create a superposition and later recombine it.

8. Global phase and relative phase

Suppose

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

Multiplying the entire state by $e^{i\gamma}$ gives

$$e^{i\gamma}|\psi\rangle.$$

This is a **global phase** and does not change physical predictions.

By contrast,

$$\alpha|0\rangle + e^{i\phi}\beta|1\rangle$$

contains a **relative phase** between the basis components. Relative phase can change the result of later gates and is therefore observable indirectly.

A practical rule:

- multiplying **all** amplitudes by the same phase: physically equivalent state;
- changing the phase of only part of the state: potentially observable.

Qiskit's `Statevector.equiv` is useful because it recognizes equality up to global phase.

9. Bloch-sphere intuition

Every pure one-qubit state can be written as

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle.$$

Ignoring global phase, the two real angles θ and ϕ specify a point on the Bloch sphere.

Important locations are:

- $|0\rangle$: north pole;
- $|1\rangle$: south pole;
- $|+\rangle$: $+x$ direction;
- $|-\rangle$: $-x$ direction.

The Bloch sphere is an intuition tool for one qubit. It does not generalize to a simple three-dimensional picture for arbitrary multi-qubit states.

10. Gate identities

Several useful identities follow from the role of H as a basis change:

$$HZH = X,$$

$$HXH = Z.$$

They will be explored in Week 4 and Week 7. The key idea is that the same physical transformation can often be represented by different gate sequences.

Circuit identities become especially important when optimizing for real hardware.

11. Prediction habit

Before running any circuit, students should be able to propagate the state by hand for simple examples.

Example:

$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{Z} |-\rangle \xrightarrow{H} |1\rangle.$$

So the circuit $H - Z - H$ starting from $|0\rangle$ must produce 1 deterministically.

The goal is not to do long matrix multiplication every time. It is to recognize common state transformations.

12. Common misconceptions

“Superposition means the qubit secretly has a classical unknown value”

A coherent superposition can generate interference effects that an ordinary classical mixture cannot.

“If two states have the same measurement histogram, they are the same state”

Only with respect to that measurement. Different bases can reveal differences.

“The minus sign in $|-\rangle$ means a negative probability”

It is a relative amplitude phase, not a probability.

“Global phase and relative phase are the same”

Global phase is physically irrelevant for isolated pure states. Relative phase can change interference.

13. What to know before the notebook

You should be able to state and use:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}},$$

$$Z|+\rangle = |-\rangle,$$

$$H^2 = I,$$

and the idea that X-basis measurement can be implemented by applying H before ordinary Z-basis measurement.

Suggested references

- IBM Quantum Learning, *Basics of Quantum Information* and current Qiskit learning material: <https://quantum.cloud.ibm.com/learning/en/courses>
- IBM Quantum Documentation and Qiskit guides: <https://quantum.cloud.ibm.com/docs>
- Quantum Rings, *Quantum 101*: <https://www.quantumrings.com/course/quantum101>
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.

The software-facing details in the practical notebook should be treated as the canonical implementation for this course. The purpose of this booklet is to explain the physics, mathematics, and algorithmic reasoning behind that implementation.